

CMAF: Continuous Masked Action-Tape Modeling for Visuomotor Manipulation

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Abstract—Action-chunking policies generate temporally coherent action sequences. We propose CMAF, a history-conditioned continuous masked action policy that extends this paradigm with selective local revision in continuous action space. CMAF treats each action chunk as an editable prediction: it predicts a coarse future chunk from the current observation and recent executed action commands, scores contiguous candidate intervals, and regenerates the selected interval while preserving the surrounding actions. CMAF achieves average success rates of 61.8% in the eight-task RoboTwin main comparison and 53.3% on ManiSkill. Real-world experiments further show consistent improvements over baseline methods on the AGIBOT G1 and Leju Kuavo 4 Pro humanoid platforms.

Index Terms—Robot Learning, Imitation Learning, Action Chunking, Masked Generation

I. INTRODUCTION

Action chunking predicts temporally coherent sequences and has proved effective across a broad range of visuomotor manipulation tasks [1], [2]. Manipulation errors often concentrate in short, task-critical intervals such as contact and alignment, motivating local revision within a predicted chunk. As illustrated in Figure 1, conventional action-chunking methods generate the future sequence as a single prediction, whereas CMAF uses executed-action history to regenerate a span while preserving surrounding actions.

Recent masked generative methods offer a natural way to move beyond this whole-chunk treatment. MGP discretizes actions into latent tokens and selectively remasks low-confidence predictions, while related discrete diffusion and masked VLA methods further explore parallel action-token generation and refinement [3], [4], [5], [6]. Continuous action-inpainting methods instead preserve committed actions and complete the remaining chunk to support asynchronous execution during closed-loop deployment [7], [8].

CMAF selectively refines native continuous action chunks. It identifies and regenerates a locally unreliable future interval while preserving the surrounding prediction. Both prediction stages are conditioned on executed-action history.

We introduce **CMAF**, a history-conditioned continuous masked action policy that treats an action chunk as an editable prediction. Given the current observation and recent executed action commands, CMAF first produces a coarse future chunk and then identifies a temporally localized region for refinement. The selected span is reopened through masking and regenerated, while the surrounding prediction is retained. This predict–select–regenerate formulation separates *where* revision is needed from *how* the selected motion should be reconstructed coherently after masking.

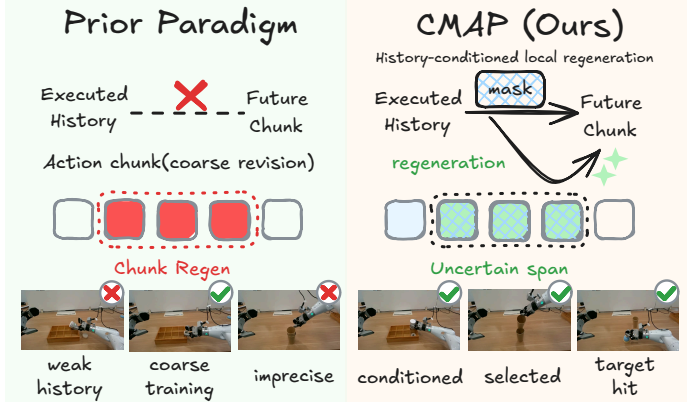


Fig. 1. Conceptual comparison between conventional action-chunk generation and CMAF. Conventional policies predict a future action chunk without explicitly revising local errors, whereas CMAF conditions prediction on executed-action history and regenerates a selected future span while preserving the remaining actions.

CMAF predicts and locally revises continuous action chunks. Recent executed commands condition future actions, and contiguous refinement regenerates a coherent temporal segment while preserving surrounding predictions. CMAF achieves average success rates of 61.8% in the eight-task RoboTwin main comparison and 53.3% on ManiSkill. Real-world experiments further show consistent gains on the AGIBOT G1 and Leju Kuavo 4 Pro platforms under contact and sensing noise.

To summarize, our main contributions are threefold:

- **CMAF framework.** A predict–select–regenerate framework that treats continuous action chunks as editable predictions for targeted local motion refinement.
- **History-conditioned masking.** A continuous action mask conditions future prediction on executed history.
- **Score-guided regeneration.** A refinement mechanism that identifies and regenerates a contiguous future interval while preserving the remaining prediction.

II. RELATED WORK

a) Visuomotor action chunking and generative policies: Imitation learning policies map visual observations and robot state to actions from demonstrations [9], [10]. Action chunking has become a strong design pattern because predicting multiple future actions jointly can improve temporal consistency and reduce the burden of single-step feedback [1], [11]. Generative policies extend this idea by modeling action chunks with richer conditional distributions: Diffusion Policy (DP) performs iterative denoising in continuous action

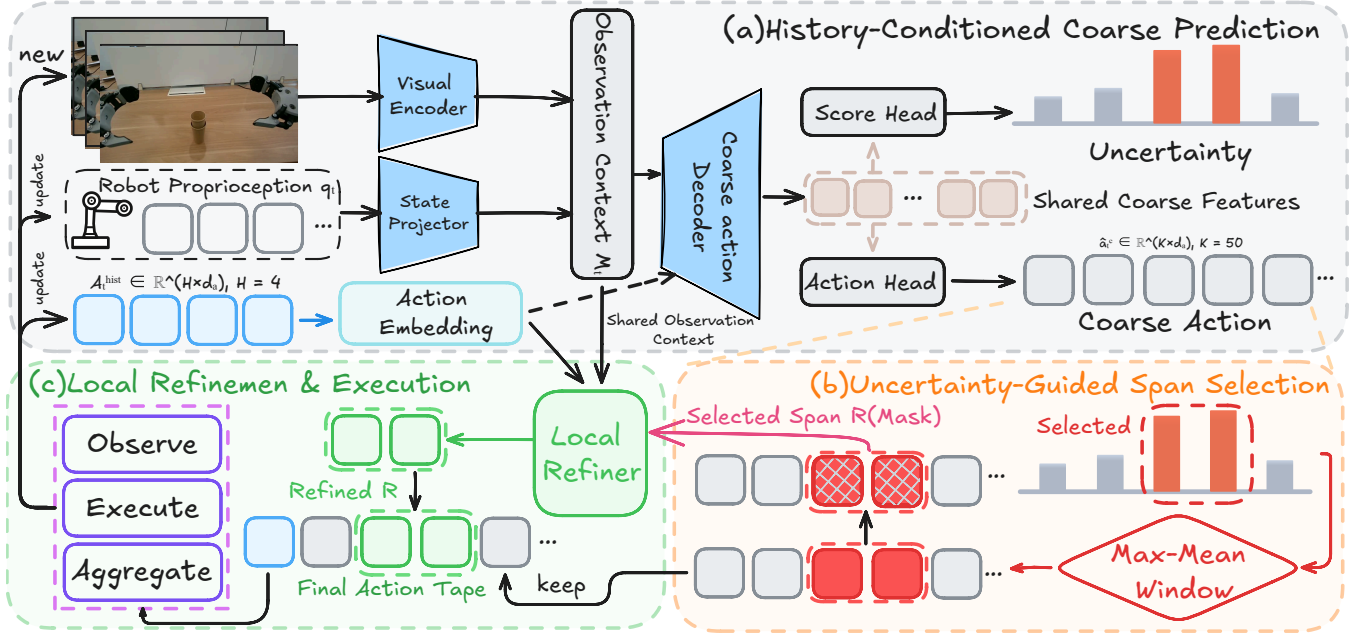


Fig. 2. Overview of CMAP. (A) The current observation and visible executed-action history of length H condition a coarse prediction of K future actions and per-step span-selection scores. (B) Contiguous future windows are ranked by their mean scores, and the highest-scoring window defines the selected span $\mathcal{R}$. (C) The local refiner regenerates only $\mathcal{R}$, merges it with the preserved coarse actions, and executes the resulting action tape with receding-horizon control.

space [2], VQ-BeT uses latent action abstractions [12], and recent large policies use diffusion or flow-style action generation for broader robot control [13], [14], [15]. These methods improve the global quality and expressiveness of action chunks. CMAP complements this direction through selective revision of local errors inside a predicted chunk.

b) Masked generation and action inpainting: Masked prediction has been widely adopted for conditional sequence completion and iterative refinement, as exemplified by Mask-Predict and MaskGIT [16], [17]. This paradigm was subsequently extended to sequential decision making through masked trajectory modeling [18], [19]. Recent robot policies further employ masked or discrete-diffusion decoding for parallel action generation and iterative refinement [3], [4], [5], [6]. Continuous action-inpainting methods use partial action conditioning for asynchronous execution [7], [8]. CMAP uses partial conditioning to regenerate a future interval while preserving unselected actions outside it during execution.

c) Selective revision and score-guided refinement: Selective refinement requires identifying which parts of a prediction are most in need of revision. Prior work has explored uncertainty estimation through heteroscedastic prediction, ensembles, and stochastic inference to characterize model confidence [20], [21], [22]. In robot learning, related reliability signals support failure detection and intervention during policy execution [23], [24], [25]. CMAP learns a score to localize a regeneration interval in each predicted chunk.

III. METHOD

Figure 2 summarizes CMAP’s predict–select–regenerate pipeline. CMAP constructs an action tape by combining H executed actions with K future action positions. It first

predicts a coarse future chunk and a per-step span-selection score v_i . For every contiguous window of length L , CMAP computes the mean score s_j and selects the maximizing window as $\mathcal{R}$. The selected span is masked and regenerated once, while all positions outside $\mathcal{R}$ retain their coarse predictions. The completed action tape is then executed with receding-horizon control during each closed-loop policy update.

A. Problem Setup

We consider visuomotor imitation learning from demonstrations. At timestep t , the policy observes visual inputs I_t , robot proprioception q_t , and a short history of executed actions. The goal is to output a future action chunk

$$\mathbf{a}_t^{\text{fut}} = [a_t, a_{t+1}, \dots, a_{t+K-1}] \in \mathbb{R}^{K \times d_a}, \quad (1)$$

where K is the chunk length and d_a is the action dimension. Let the recent action history be

$$\mathbf{a}_t^{\text{hist}} = [a_{t-H}, \dots, a_{t-1}] \in \mathbb{R}^{H \times d_a}. \quad (2)$$

The full CMAP instantiation concatenates the two parts into an action tape

$$\mathbf{x}_t = [\mathbf{a}_t^{\text{hist}} \mid \mathbf{a}_t^{\text{fut}}] \in \mathbb{R}^{(H+K) \times d_a}. \quad (3)$$

The tape is a conditional prediction object: visible history positions provide context, while masked future positions define the policy query used to predict future actions.

We denote a binary mask by $\mathbf{m} \in \{0, 1\}^{H+K}$, where $m_i = 1$ indicates a hidden position. The main policy query keeps all valid history positions visible and masks all valid future positions. ManiSkill instead projects the last H actions linearly into the state embedding; both benchmarks retain the same future-chunk prediction and span-regeneration stages.

TABLE I

ROBOTWIN TASK SUCCESS RATES (%). AVG. IS THE EIGHT-TASK MACRO-AVERAGE. BEST VALUES AMONG ACT, DP, AND CMAP ARE BOLD, AND CMAP IS SHADED.

Method	Beat Block Hammer	Pick Dual Bottles	Handover Mic	Stack Two Blocks	Open Microwave	Click Alarm Clock	Open Laptop	Place Empty Cup	Avg.
RDT-1B	77.0	42.0	90.0	21.0	37.0	61.0	59.0	56.0	55.4
TwinVLA [26]	77.0	18.0	84.0	26.0	3.0	33.0	80.0	50.0	46.4
ACT	45.0	20.0	83.0	19.0	65.0	29.0	60.0	45.0	45.8
SeedPolicy-T [27]	72.0	24.0	92.0	47.0	80.0	61.0	45.0	32.0	56.6
Diffusion Policy (DP)	42.0	24.0	53.0	7.0	5.0	61.0	49.0	37.0	34.8
CMAP (Ours)	65.0	26.0	88.0	23.0	92.0	72.0	72.0	56.0	61.8

TABLE II

MANISKILL SUCCESS RATES (%) AND MACRO-AVERAGE. BOLD COMPARES ACT AND CMAP; CMAP IS SHADED, AND † MARKS RESULTS REPORTED BY STAMO [28].

Method	Push Cube	Stack Cube	Pull Cube	Avg.
OpenVLA†	4.0	18.0	16.0	12.7
UniVLA†	7.0	22.0	20.0	16.3
WorldVLA†	10.0	17.0	22.0	16.3
ACT	31.0	13.0	34.0	26.0
StaMo†	32.0	16.0	38.0	28.7
OpenVLA + StaMo state†	32.0	48.0	36.0	38.7
DINO-WM†	8.0	4.0	18.0	10.0
CMAP (Ours)	54.0	18.0	88.0	53.3

TABLE III

CMAP SENSITIVITY IN PAIRED FOUR-TASK ROBOTWIN PILOTS (%). ρ , P , AND L DENOTE CONTEXT MASKING, REGENERATION PASSES, AND SPAN LENGTH; THE BETTER RESULT IN EACH PAIR IS BOLD.

Setting	Value	Success
Context mask ρ (RoboTwin)	0.0 (default) 0.3	73.0 69.8
Regeneration passes P (RoboTwin)	1 (default) 2 (+1 pass)	59.5 60.0
Span-length schedule (RoboTwin)	Fixed $L = 12$ (default) Adaptive $L \in \{8, 12, 16\}$	59.5 58.8

B. Continuous Masked Action-Tape Modeling

CMAP embeds each RoboTwin tape position with a position query and either a visible action embedding or a learned mask embedding. For a visible action x_i , the input token is

$$\mathbf{z}_i = \mathbf{e}_i + \phi_a(x_i), \quad (4)$$

where $\mathbf{e}_i$ is a learned positional query and ϕ_a projects continuous actions into the model dimension. For a masked position, the input token is

$$\mathbf{z}_i = \mathbf{e}_i + \mathbf{e}_{\text{mask}}. \quad (5)$$

A visual encoder extracts observation features from I_t , proprioceptive features are computed from q_t , and a transformer predicts continuous future actions conditioned on these features and the available history representation. A latent encoder produces $q_\phi(z | \cdot) = \mathcal{N}(\mu_z, \text{diag}(\sigma_z^2))$. In the RoboTwin implementation, the latent encoder receives visible history and zero-filled future positions. The regularizer is

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_j (1 + \log \sigma_{z,j}^2 - \mu_{z,j}^2 - \sigma_{z,j}^2), \quad (6)$$

At inference, CMAP sets the latent variable to zero.

C. Masked Training and Objective

Training deterministically masks all valid future positions and keeps history visible in the main configuration. Let $\hat{\mathbf{a}}^c$ be the coarse chunk, $\hat{\mathbf{a}}^r$ the regenerated prediction on selected

span $\mathcal{R}$, and $\hat{\mathbf{a}}^f$ the final chunk after replacement. For N valid future steps, the losses are

$$\begin{aligned} \mathcal{L}_{\text{coarse}} &= \frac{1}{N} \sum_i \|\mathbf{a}_i - \hat{\mathbf{a}}_i^c\|_1, \\ \mathcal{L}_{\text{regen}} &= \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} \|\mathbf{a}_i - \hat{\mathbf{a}}_i^r\|_1, \\ \mathcal{L}_{\text{final}} &= \frac{1}{N} \sum_i \|\mathbf{a}_i - \hat{\mathbf{a}}_i^f\|_1. \end{aligned} \quad (7)$$

The losses are normalized over valid time steps, and each ℓ_1 norm sums across all action dimensions. RoboTwin predicts a clipped log-variance $\ell_i \in [-5, 5]$ from a stop-gradient decoder state and supervises it with the stop-gradient coarse squared error,

$$e_i^{\text{RT}} = \frac{1}{d_a} \|\mathbf{a}_i - \text{sg}(\hat{\mathbf{a}}_i^c)\|_2^2, \quad \mathcal{L}_{\text{unc}}^{\text{RT}} = \frac{1}{N} \sum_i [e^{-\ell_i} e_i^{\text{RT}} + \ell_i]. \quad (8)$$

We use this quantity to score candidate refinement spans. ManiSkill predicts $u_i = \text{softplus}(Wh_i)$ and regresses the normalized coarse ℓ_1 error,

$$e_i^{\text{MS}} = \frac{1}{d_a} \|\text{sg}(\hat{\mathbf{a}}_i^c) - \mathbf{a}_i\|_1, \quad \mathcal{L}_{\text{unc}}^{\text{MS}} = \text{MSE}(u_i, e_i^{\text{MS}}). \quad (9)$$

The benchmark-specific objectives are

$$\begin{aligned} \mathcal{L}_{\text{RT}} &= \mathcal{L}_{\text{coarse}} + \mathcal{L}_{\text{regen}} + \mathcal{L}_{\text{final}} + \mathcal{L}_{\text{unc}}^{\text{RT}} + 10\mathcal{L}_{\text{KL}}, \\ \mathcal{L}_{\text{MS}} &= \frac{1}{3}(\mathcal{L}_{\text{coarse}} + \mathcal{L}_{\text{regen}} + \mathcal{L}_{\text{final}}) + 0.1\mathcal{L}_{\text{unc}}^{\text{MS}} + 10\mathcal{L}_{\text{KL}}. \end{aligned} \quad (10)$$

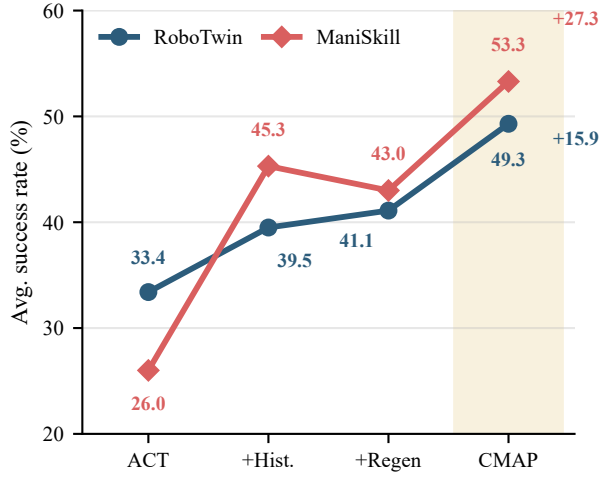


Fig. 3. Component ablations on the fixed RoboTwin diagnostic suite and ManiSkill. +Hist. adds executed-action history, +Regen adds masked span regeneration, and CMAP adds score-guided span selection; labels report macro-averages and CMAP gains over ACT.

D. Selective Span Regeneration

After coarse prediction, CMAP assigns each future position a span-selection score v_i , using $v_i = \ell_i$ on RoboTwin and $v_i = u_i$ on ManiSkill. For span length L , it scores every contiguous candidate window by

$$s_j = \frac{1}{L} \sum_{i=j}^{j+L-1} v_i, \quad j^* = \arg \max_j s_j. \quad (11)$$

The first maximizing window is retained when scores tie. CMAP masks $\mathcal{R} = \{j^*, \dots, j^* + L - 1\}$, performs one regeneration pass, and replaces only the selected positions:

$$\hat{a}_i^f = \begin{cases} \hat{a}_i^r, & i \in \mathcal{R}, \\ \hat{a}_i^c, & i \notin \mathcal{R}. \end{cases} \quad (12)$$

Positions outside $\mathcal{R}$ remain unchanged; all reported models select one span and apply one regeneration pass.

IV. SIMULATION EXPERIMENTS

A. Benchmarks and Evaluation Protocols

We evaluate CMAP on eight RoboTwin tasks [29] and three single-arm ManiSkill tasks [30], using 50 and 100 expert demonstrations per task, respectively. The RoboTwin main-comparison tasks shown in Figure 5 are *Beat Block Hammer*, *Pick Dual Bottles*, *Handover Mic*, *Stack Two Blocks*, *Open Microwave*, *Click Alarm Clock*, *Open Laptop*, and *Place Empty Cup*. Together, they cover tool-mediated contact, simultaneous dual-arm pickup, cross-arm handover, stacking and alignment, articulated-object manipulation, contact triggering, and goal-directed placement. The bottom row of Figure 5 shows frames recorded directly from CMAP evaluation rollouts on ManiSkill. During execution, the policy first moves the gripper into task-relevant contact, then steers the cube toward the marked goal in *Push Cube* and *Pull Cube*, or approaches and manipulates the upper cube for grasping and alignment in *Stack Cube*. RoboTwin

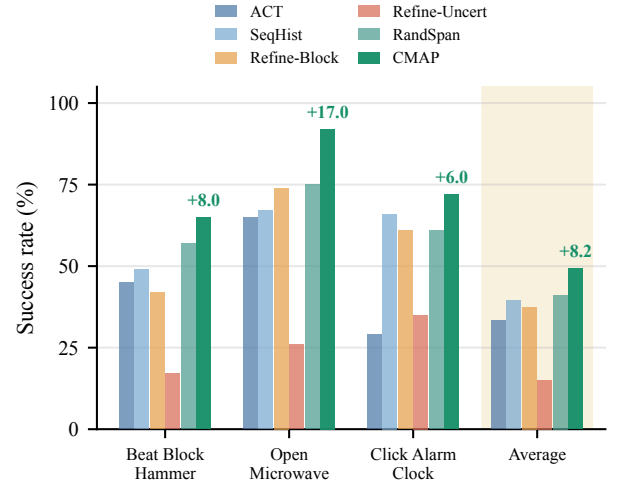


Fig. 4. Task-level ablations on three tasks from the fixed RoboTwin diagnostic suite and its eight-task macro-average. Labels above CMAP show gains over the strongest alternative.

baselines include ACT [1], Diffusion Policy (DP), RDT-1B [14], TwinVLA [26], and SeedPolicy-T [27]. ACT, DP, and CMAP are each evaluated over 100 episodes per task, and Table I reports the task-wise success rates and eight-task macro-averages; the other rows retain their published point estimates for the same benchmark tasks.

B. Performance Comparison

a) *RoboTwin Main Comparison:* Table I summarizes the RoboTwin results. Among ACT, DP, and CMAP, CMAP achieves the highest success rate on all eight tasks and raises the macro-average to 61.8%, compared with 45.8% for ACT and 34.8% for DP. It also exceeds the strongest published macro-average in the table, 56.6% from SeedPolicy-T. CMAP reaches 88.0% on *Handover Mic* and 56.0% on *Place Empty Cup*, retaining strong performance on coordinated transfer and goal-directed placement.

b) *ManiSkill Comparison:* Table II summarizes the ManiSkill results. Boldface compares ACT and CMAP; CMAP is higher on all three tasks and the overall average. CMAP records 54.0%, 18.0%, and 88.0% on *Push Cube*, *Stack Cube*, and *Pull Cube*, respectively, averaging 53.3%.

C. Ablation Studies

a) *Core Components:* Figure 3 compares four matched configurations across RoboTwin and ManiSkill. ACT is the base policy; SeqHist adds executed-action history; RandSpan adds contiguous masked regeneration with random span placement; and CMAP replaces random placement with score-guided span selection. All RoboTwin component ablations use a fixed eight-task diagnostic suite: *Beat Block Hammer*, *Pick Dual Bottles*, *Hanging Mug*, *Stack Two Blocks*, *Open Microwave*, *Click Alarm Clock*, *Open Laptop*, and *Turn Switch*. Training and evaluation settings are matched across configurations within this suite.

Figure 4 expands the diagnostic-suite comparison to six configurations on three representative tasks and the eight-task

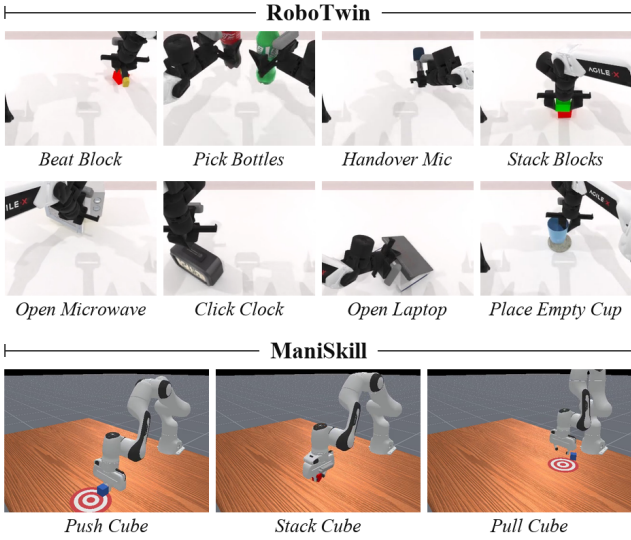


Fig. 5. Simulation tasks in RoboTwin (top two rows) and ManiSkill (bottom row). ManiSkill frames are sampled from CMAP evaluation rollouts.

macro-average. Refine-Block performs residual refinement over a contiguous block, while Refine-Uncert refines individually selected high-uncertainty steps. RandSpan and CMAP instead regenerate masked spans using random and score-guided selection, respectively. CMAP annotations show gains over the best alternative in every reported group.

b) Configuration Sensitivity: Table III reports three paired four-task RoboTwin pilots. The context-mask pair uses *Beat Block Hammer*, *Open Microwave*, *Open Laptop*, and *Click Alarm Clock*; the pass-count and span-schedule pairs use *Beat Block Hammer*, *Open Microwave*, *Click Alarm Clock*, and *Stack Two Blocks*. Each pair holds its task subset and training budget fixed. Visible history ($\rho = 0$) improves success by 3.2 points, a second regeneration pass changes success by 0.5 points, and fixed and adaptive spans obtain 59.5% and 58.8%, respectively.

c) Cross-Benchmark Ablation Summary: Figure 3 compares the four matched internal configurations on the fixed RoboTwin diagnostic suite and on ManiSkill. Relative to ACT, the history-conditioned configuration improves the average success rate from 33.4% to 39.5% on the RoboTwin diagnostic suite and from 26.0% to 45.3% on ManiSkill. RandSpan obtains 41.1% and 43.0%, respectively. Replacing random span placement with score-guided selection raises the averages to 49.3% on the RoboTwin diagnostic suite and 53.3% on ManiSkill. Per prediction cycle, CMAP regenerates 12 of 50 future actions on RoboTwin and 8 of 30 on ManiSkill during deployment.

V. REAL-WORLD EXPERIMENTS

A. Real Robot Benchmark

We evaluate ACT, DP, and CMAP on the AGIBOT G1 and Leju Kuavo 4 Pro platforms. The AGIBOT setup controls the right 7-DoF arm and scalar gripper from head- and right-wrist-camera RGB streams, with a 15-Hz policy loop and a 100-Hz low-level arm-control loop. The Kuavo setup

controls one 7-DoF arm equipped with a dexterous hand from synchronized multi-view RGB observations at 10 Hz. Together, these physical deployments expose the policies to contact dynamics, arm-hand coordination, sensing noise, control latency, and communication delay. Figures 6 and 7 present representative rollouts on Kuavo 4 Pro and AGIBOT G1, respectively. Table IV reports the corresponding results under this common evaluation protocol for both platforms.

a) Kuavo 4 Pro Tasks: The Kuavo 4 Pro evaluation focuses on one long-horizon, single-arm dexterous-manipulation task, *Sheet Loading*. As shown in Figure 6, the four displayed stages cover the initial configuration, pre-grasp alignment, grasp and lift, and placement and release. During the transition between the last two frames, the robot transports the sheet-metal workpiece toward the target fixture while maintaining a stable grasp. Completing the sequence requires coordinated arm-hand control to grasp, transport, align, and release the workpiece at the target fixture.

b) AGIBOT G1 Tasks: The AGIBOT G1 suite contains three single-arm tabletop tasks: *Cup Separation*, which removes the upper cup from two nested disposable paper cups; *Bell Press*, which activates a service-bell-style button; and *Dual-Item Placement*, whose objective is to place one chess piece and one small bottle into separate compartments of an open box. For the last task, we report two predefined sequential checkpoints, *Chess Grasp* and *Bottle Grasp*. These tasks evaluate short contact transitions, precise triggering, and progression through a sequential multi-object interaction. Figure 7 visualizes representative rollouts for the three tasks: columns correspond to *Cup Separation*, *Bell Press*, and *Dual-Item Placement*, while rows trace successive execution stages from approach to task completion for each rollout.

B. Deployment Protocol

We collect 50 teleoperated demonstrations per task and use the same task-specific demonstration set to train ACT, DP, and CMAP. The deployment logging pipelines record synchronized multi-view RGB streams, proprioception, action commands, and timestamps. The Kuavo sequence in Figure 6 visualizes four representative stages sampled from one CMAP physical-evaluation rollout. Each method is evaluated over 10 physical trials per task with matched observations and action spaces. Success outcomes are manually verified from the recorded videos and logs. *Chess Grasp* and *Bottle Grasp* are scored as paired checkpoints within the same 10 *Dual-Item Placement* trials evaluated for each method.

Both platforms enforce joint/workspace limits, joint-delta clipping, emergency stops, and home-reset procedures throughout every recorded physical trial on both platforms.

C. Performance Comparison

Table IV reports success counts and rates for all three methods. On AGIBOT G1, the aggregate rates over the four reported task or checkpoint outcomes are 57.5% for ACT, 57.5% for DP, and 85.0% for CMAP. On Kuavo 4 Pro *Sheet Loading*, ACT, DP, and CMAP obtain 2/10, 1/10, and 4/10, respectively. Aggregating all five outcomes gives 25/50

TABLE IV

REAL-WORLD SUCCESS COUNTS AND RATES ON AGIBOT G1 AND KUAVO 4 PRO. *Chess Grasp* AND *Bottle Grasp* ARE CHECKPOINTS WITHIN 10 SHARED *Dual-Item Placement* TRIALS; OUTCOME AVG. AGGREGATES FIVE OUTCOMES. BEST VALUES ARE BOLD, AND CMAP IS SHADED.

Method	AGIBOT G1				Kuavo 4 Pro	Outcome Avg.
	Cup Separation	Bell Press	Dual-Item Placement		Sheet Loading	
			Chess Grasp	Bottle Grasp		
ACT	4/10 (40.0)	8/10 (80.0)	5/10 (50.0)	6/10 (60.0)	2/10 (20.0)	25/50 (50.0)
DP	5/10 (50.0)	7/10 (70.0)	3/10 (30.0)	8/10 (80.0)	1/10 (10.0)	24/50 (48.0)
CMAP (Ours)	7/10 (70.0)	9/10 (90.0)	8/10 (80.0)	10/10 (100.0)	4/10 (40.0)	38/50 (76.0)

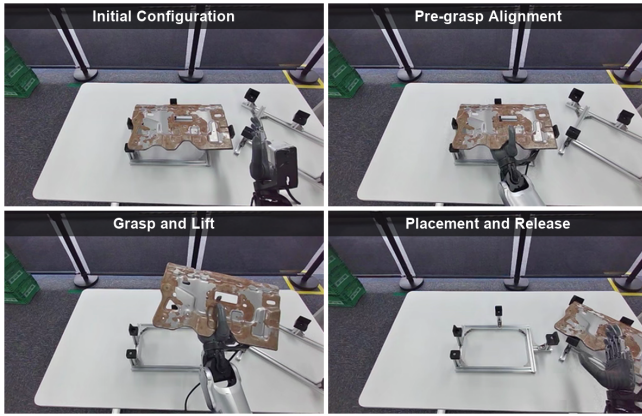


Fig. 6. Four stages of a CMAP rollout for Kuavo 4 Pro *Sheet Loading*: initialization, alignment, grasp and lift, and placement and release.

(50.0%) for ACT, 24/50 (48.0%) for DP, and 38/50 (76.0%) for CMAP. The 50 matched outcomes combine four AGIBOT task or checkpoint measures with Kuavo *Sheet Loading*.

a) Cross-Platform Trends: CMAP improves the aggregate AGIBOT G1 success rate by 27.5 percentage points over both ACT and DP. The gain is distributed across the evaluation suite: CMAP reaches 70.0% on *Cup Separation*, 90.0% on *Bell Press*, 80.0% on *Chess Grasp*, and 100.0% on *Bottle Grasp*. The corresponding improvements over ACT are 30, 10, 30, and 40 percentage points. Relative to DP, the gains are 20, 20, 50, and 20 points. On Kuavo 4 Pro, CMAP completes 4 of 10 *Sheet Loading* trials, compared with 2 for ACT and 1 for DP. The overall improvement therefore reflects consistent gains across two robot embodiments rather than a single dominant outcome.

b) Execution-Level Interpretation: The physical tasks expose complementary demands on online correction. *Cup Separation* and *Bell Press* concentrate the decisive motion near a short contact transition, whereas *Dual-Item Placement* requires the policy to preserve progress across two ordered object interactions. *Sheet Loading* further tests whether a stable grasp can be maintained during a longer transport-and-alignment sequence. CMAP uses the executed-action history to anchor each new prediction to the motion already issued by the controller, then limits regeneration to one contiguous future interval. This design preserves unaffected phases of the coarse action tape while focusing model capacity on the segment selected for revision. The results indicate that the same refinement mechanism

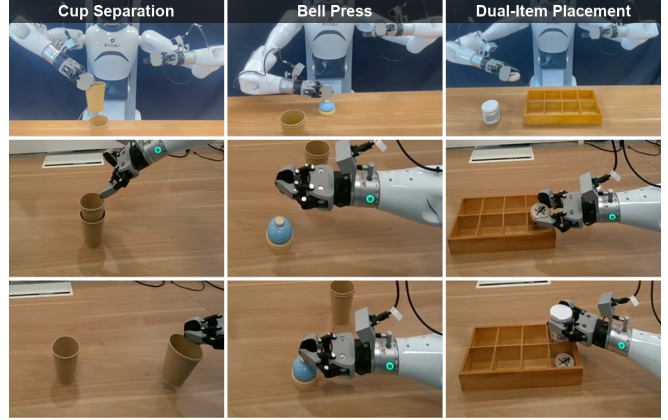


Fig. 7. Representative AGIBOT G1 task rollouts. Columns denote tasks, and rows show successive execution stages.

remains useful across short triggering motions, sequential grasping, and long-horizon transport.

c) Representative Execution Cases: Figure 8 pairs successful outcomes with four contact failures and one recovered near-failure on AGIBOT G1 and Kuavo 4 Pro. The columns show *Cup Separation*, *Bell Press*, *Chess Grasp*, *Bottle Grasp*, and *Sheet Loading*. Red callouts mark the critical contact regions.

Cup Separation misses rim contact, while *Bell Press* applies force off center. During *Dual-Item Placement*, incomplete closure on the chess piece and a tilted bottle grasp undermine placement. *Sheet Loading* briefly relies on single-point support before recovering. These cases identify contact geometry, grasp closure, and object alignment as the main real-world failure sources.

The current policy scores future actions from visual observations, proprioception, and executed commands, without direct tactile or force measurements. The remaining failures identify a concrete extension path: contact sensing can augment the span-selection score, while a recovery controller can reopen a new interval when the expected contact event is absent. This combination would preserve CMAP’s localized editing structure while improving detection and correction of contact-specific errors during closed-loop execution.

D. Discussion

a) Ablation-to-Deployment Connection: The simulation ablations clarify how CMAP’s components contribute to the final policy. On the fixed RoboTwin diagnostic suite,

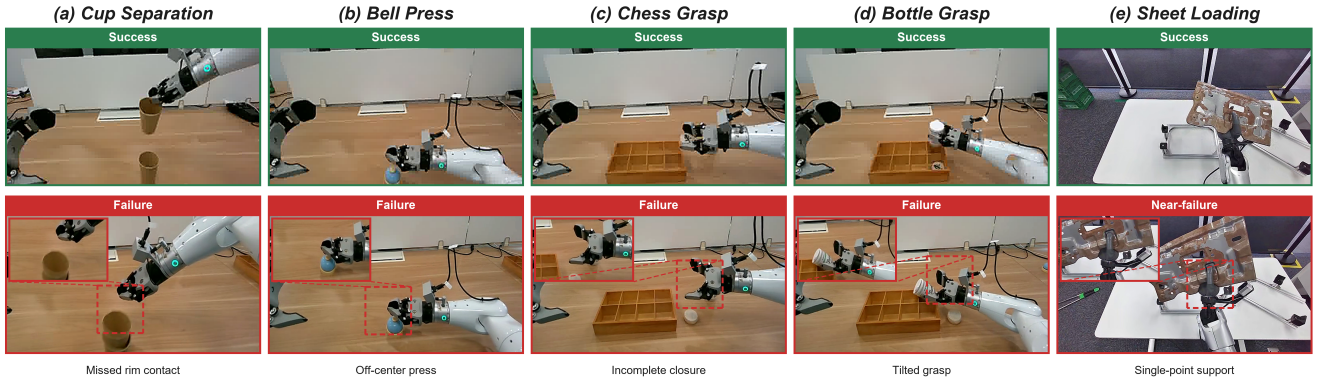


Fig. 8. Real-world execution examples on AGIBOT G1 and Kuavo 4 Pro. Top: success. Bottom: failures for *Cup Separation*, *Bell Press*, *Chess Grasp*, and *Bottle Grasp*, plus a recovered near-failure for *Sheet Loading*. Red callouts mark critical contact regions.

adding executed-action history raises average success from 33.4% to 39.5%, and random-span regeneration reaches 41.1%; the corresponding ManiSkill values are 26.0%, 45.3%, and 43.0%. Replacing random placement with score-guided selection increases the averages to 49.3% on the diagnostic suite and 53.3% on ManiSkill. The same ordering is reflected in the task-level diagnostic comparison, where CMAP exceeds the strongest alternative in all four reported groups. Together, these results connect the two central design decisions: executed history anchors the new prediction to realized control, and span scoring directs refinement toward the portion of the future tape most likely to benefit from revision.

b) Temporal Editing Granularity: CMAP deliberately edits a minority of each predicted chunk. During deployment it regenerates 12 of 50 future actions on RoboTwin and 8 of 30 on ManiSkill, corresponding to 24.0% and 26.7% of the respective horizons. The remaining positions preserve the coarse prediction and therefore retain the global structure of the planned motion. In the paired four-task pilots of Table III, a second regeneration pass changes success by 0.5 points, while adaptive span scheduling reaches 58.8% compared with 59.5% for the fixed-span setting. These results support a conservative online update: one contiguous edit captures most of the observed benefit without repeatedly rewriting the complete action chunk.

c) Task-Dependent Behavior: The task-level results also identify where local regeneration is most effective. In the RoboTwin main comparison, CMAP leads the matched ACT and DP results on all eight tasks. Its gains over the stronger matched baseline are 27 points on *Open Microwave*, 20 on *Beat Block Hammer*, 12 on *Open Laptop*, and 11 on both *Click Alarm Clock* and *Place Empty Cup*. The smaller gains on *Pick Dual Bottles* and *Stack Two Blocks*, at 2 and 4 points, show that precision-heavy coordination remains demanding. On ManiSkill, the gains over ACT are 23 points on *Push Cube*, 5 points on *Stack Cube*, and 54 points on *Pull Cube*. The variation across tasks indicates that score-guided regeneration is especially valuable when a localized correction can recover the remainder of a motion.

d) Evaluation Scope and Extensions: The physical study uses matched training data and action spaces, 50 teleoperated demonstrations per task, and 10 trials for each reported task or checkpoint. This protocol isolates the policy comparison on the two robot platforms and exposes the methods to distinct camera layouts, control frequencies, end effectors, and contact patterns. A broader evaluation can extend this evidence through additional object instances, scene layouts, lighting conditions, and repeated training seeds. The diagnostic pairs further motivate contact-sensitive observations, adaptive span lengths, and event-triggered recovery passes. These extensions fit the existing action-tape formulation: new sensing channels can inform the score head, while the selected interval remains the unit of localized regeneration and closed-loop correction.

VI. CONCLUSION

We presented CMAP, a history-conditioned continuous masked action-tape framework for local revision of predicted action chunks. CMAP conditions coarse future prediction and local regeneration on recently executed actions, scores contiguous candidate spans, and applies one regeneration pass to the highest-scoring span while preserving the remaining coarse prediction. CMAP obtains average success rates of 61.8% in the eight-task RoboTwin main comparison and 53.3% on ManiSkill. Controlled experiments on the fixed RoboTwin diagnostic suite and ManiSkill distinguish the contributions of visible action history, contiguous masked regeneration, and score-guided span selection. Across the five reported physical outcomes, CMAP achieves 38 successes in 50 trials, compared with 25 for ACT and 24 for DP. The task-level results show that localized regeneration benefits both short contact events and longer transport sequences, while the paired examples isolate contact establishment as the main remaining challenge. Future work will calibrate span-selection scores, incorporate contact-sensitive observations, adapt regeneration budgets, and broaden physical evaluation across additional robot platforms, task families, and contact conditions.

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